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AI-ENHANCED DFIR: “ThreatLens AI” FOR VELOCIRAPTOR

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ABSTRACT

As cyber threats continue to evolve in complexity and speed, the ability to detect, investigate, and respond to incidents efficiently has become a critical requirement in modern cybersecurity operations. This capstone project aims to enhance the open-source digital forensics and incident response platform Velociraptor by integrating Large Language Model (LLM) capabilities to streamline investigation workflows and improve decision support during incident response.

OBJECTIVE

1. Automate Velociraptor’s Artifact collection and Analysis
2. Train a domain-specific LLM using Velociraptor.
3. Maintain privacy and accountability.
4. Reduce analyst workload and investigation time

BENIFITS

- Operational Efficiency: Automates repetitive DFIR tasks, cutting investigation turnaround time.
- Enhanced Accessibility: Enables analysts of varying skill levels to perform complex forensic tasks easily.
- Data Privacy: Runs entirely on-premises with no cloud dependency, ensuring confidentiality.



METHODOLOGY

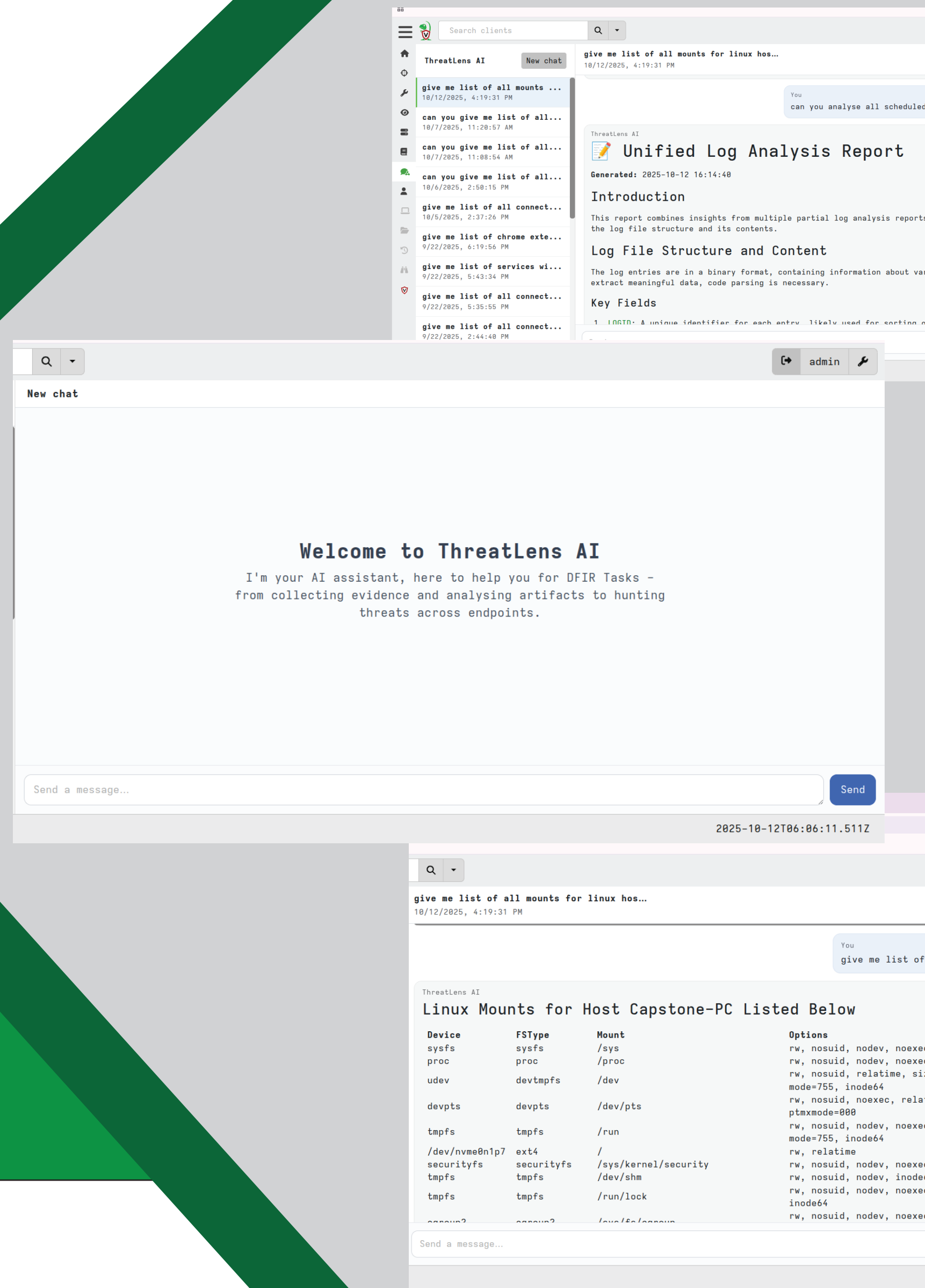
Phase 1: Integrate a Locally hosted LLM with Velociraptor, using Tools server and API. Create Chat-based interface for user prompt conversation.

Phase 2: Fine-tune the LLM on official documentation and synthetic logs for contextual assistance and analytical reasoning.



OUTCOMES

The resulting prototype delivers an integrated AI assistant within Velociraptor capable of executing and analysing forensic artefact collections through natural language. It demonstrates improved efficiency, analyst guidance, and local data governance.



Connect with us!



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“The truth is in the logs. Let’s dig deeper!”